School of Computer Science and Engineering

B. Tech (Computer Science and Engineering)

WIN Semester 21-22

SWE3001 – FUNDAMENTALS OF COMPUTER NETWORKS

Lab Manual

2021-22

Grading Policy

Experiment No.	Marks	Weightage
1	10	25%
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
8	10	
9	10	
10	10	

Software Requirement

Cisco Packet Tracer

Official Site:

https://www.netacad.com/courses/packet-tracer

Download Link:

https://www.computernetworkingnotes.com/ccna-study-guide/download-packet-tracer-for-windows-and-linux.html

Operating System: Windows

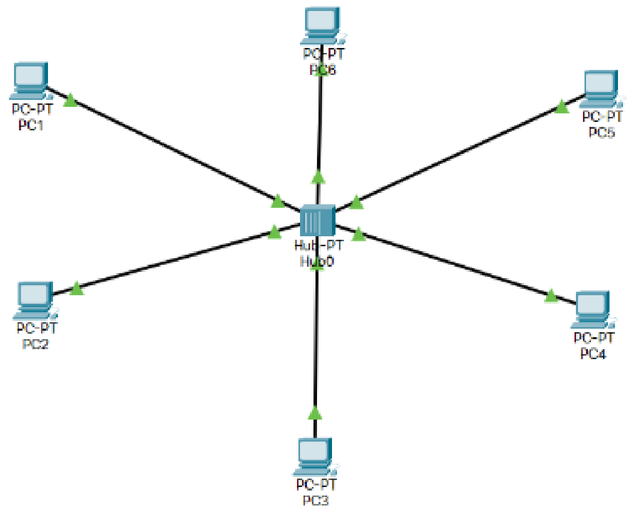
Lab Exercises

SL. No	Title
1	Simulation of LAN using HUB
2	Design a LAN/MAN network using a Single Switch
3	Creating and configuring VLANs
4	Design of Topologies: Bus, Mesh, Ring, Tree and Star topologies
5	Port-Security Configuration to prevent MAC flooding attack
6	Router Serial Point to Point Connection HDLC, PPP with PAP and CHAP
7	Design a network using Subnetting for a class C Ipv4 address and Configure it in Router.
8	Design and configure a network using RIP Routing protocol.
9	Design and configure a network using OSPF Routing protocol.

Lab 1: Simulation of LAN using the hub

Objectives:

1) Basics of Cisco Packet Tracer.
2) Hub.
3) Cisco Packet Tracer Simulation of LAN using the hub.
4) Creating a LAN using the hub.
5) Pros and Cons of the hub



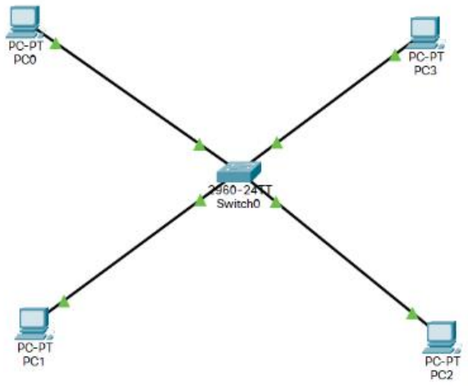
<u>Process:</u>

- Select a hub with 6 pc's and connect them through a wire which is of Straight away connection LAN
- Select each of the system and keep a specific ip-address by changing their last no. Example (10.10.10.1, 10.10.10.2, 10.10.10.3, 10.10.10.4 etc.)
- Now select the packet in the packet tracer and select the sender and receiver.
- Now Simulate the corresponding file
- Also we can create many Scenarios by changing the sender and receiver.

Lab 2: a) Design a LAN network using a Single Switch

Objectives:

1) Design a LAN using a switch with four PCs.
2) Verify the connectivity.



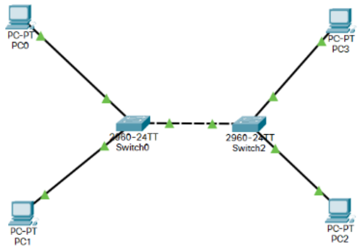
Addressing Table:

Device	Interface	IP Address	Subnet Mask
PC1	NIC	10.10.10.1	255.0.0.0
PC2	NIC	10.10.10.2	255.0.0.0
PC3	NIC	10.10.10.3	255.0.0.0
PC4	NIC	10.10.10.4	255.0.0.0

<u>Procedure:</u>

- Take 4 Pc's and a switch and connect them with a copper straight-Through as Shown.
- Now configure each pc with their ip address as pc0-10.10.10.1, pc1-10.10.10.2, pc2-10.10.10.3, pc4-10.10.10.4.
- Now send a packet from onside of the switch to the other side of the switch.
- Now run the simulation.

b) Design a LAN network using a Two Switches.



Objectives:

1. Design a LAN using two switches with four PCs.
2. Verify the connectivity.

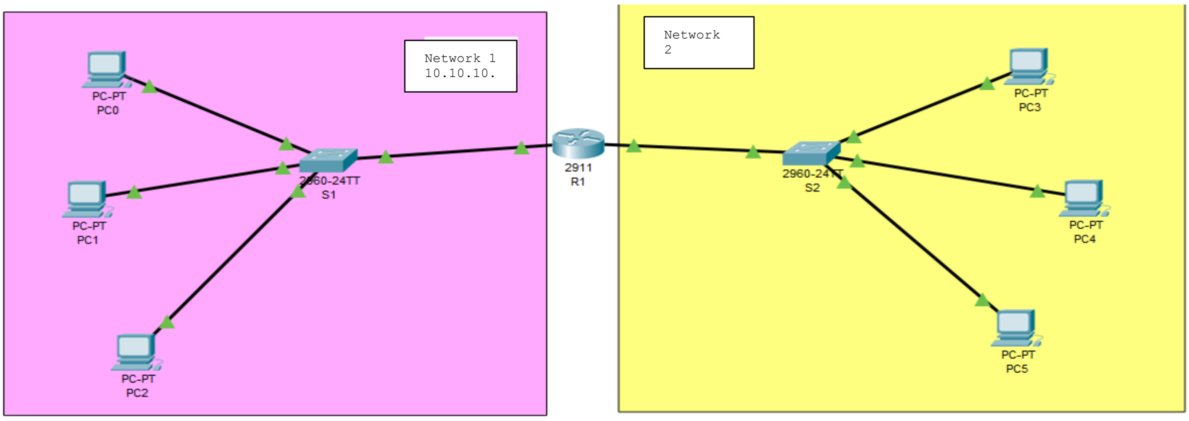
Addressing Table:

Device	Interface	IP Address	Subnet Mask
PC1	NIC	10.10.10.1	255.0.0.0
PC2	NIC	10.10.10.2	255.0.0.0
PC3	NIC	10.10.10.3	255.0.0.0
PC4	NIC	10.10.10.4	255.0.0.0

<u>Procedure:</u>

- Take 4 Pc's and two switches and connect them with a copper straight-Through and then connect the both Switches with Copper cross-Over.
- Now configure the each pc with their ip address as pc0-10.10.10.1, pc1-10.10.10.2, pc2-10.10.10.3, pc4-10.10.10.4.
- Now send a packet from onside of the switch to the other side of the switch.
- Now run the simulation.

c) Design a MAN network using a Single Router and Configure Router with CLI Mode.



Objectives:

1) Design two LAN using two switches with three PCs each.
2) Add one Router to connect two LAN networks
3) Verify the connectivity.

Addressing Table:

Device	Interface	IP Address	Subnet Mask
PC0	NIC	10.10.10.2	255.0.0.0
PC1	NIC	10.10.10.3	255.0.0.0
PC2	NIC	10.10.10.4	255.0.0.0
PC3	NIC	11.11.11.2	255.0.0.0
PC4	NIC	11.11.11.3	255.0.0.0
PC5	NIC	11.11.11.4	255.0.0.0
Router 1	NIC	10.10.10.1	255.0.0.0
Router 1 2nd int	INC	11.11.11.1	255.0.0.0
LAN 1		10.10.10.0	255.0.0.0
LAN 2		11.11.11.0	255.0.0.0

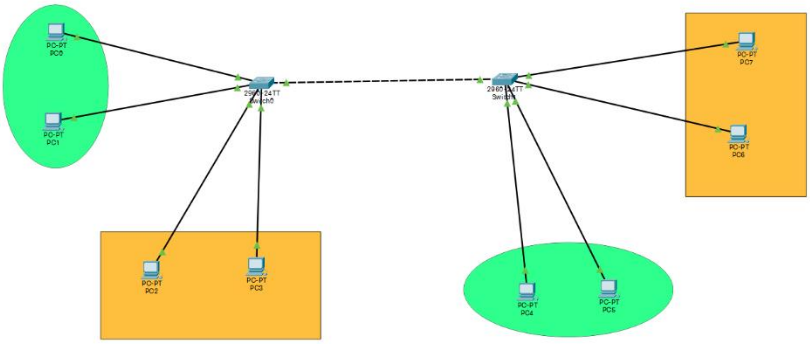
Procedure:

- Take a set of 6 pc's and separate them with 3 pc with one class of IP and 3 pc's with another class of IP
- Now we use a switch two connect each of the LAN series as shown in figure.
- Now we connect the both switches to the Router. Now the Router take them as two different gateways.
- Here I have given pc0-10.10.10.2, pc1-10.10.10.3, pc2-10.10.10.4 we are keeping 10.10.10.1 a side for the router configuration as this is the LAN1
- Now for LAN2 we going in the same process by keeping anther series of IP address pc3-11.11.11.2, pc4-11.11.11.3, pc5-11.11.11.4 we are keeping 11.11.11.1 aside for the router configuration as this is the LAN2
- Now we need to configure the router as shown above.
- Now our connection is done. Lets send a packet to confirm that our connection is successful or not.

Lab 3: Creating and configuring 2 VLANs

Objectives:

1) Verify the default VLAN configuration
2) Configure the VLANs
3) Assign VLANs to ports.



Addressing Table:

Device	Interface	IP Address	Subnet Mask	VLAN
PC0	NIC	10.10.10.1	255.0.0.0	10
PC1	NIC	10.10.10.2	255.0.0.0	10
PC2	NIC	10.10.20.1	255.0.0.0	20
PC3	NIC	10.10.20.2	255.0.0.0	20
PC4	NIC	10.10.10.3	255.0.0.0	10
PC5	NIC	10.10.10.4	255.0.0.0	10
PC6	NIC	10.10.20.3	255.0.0.0	20
PC7	INC	10.10.20.4	255.0.0.0	20

1. Verify the default VLAN configuration

2. Configure the VLANs

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram, and cable as necessary.

Step 2: Initialize and reload the switches as necessary.

Step 3: Configure basic settings for each switch.

- Console into the switch and enter global configuration mode.
- Copy the following basic configuration and paste it to the running-configuration on the switch.
- Configure the host name as shown in the topology.
- Configure the IP address listed in the Addressing Table for VLAN 1 on the switch.
- Administratively deactivate all unused ports on the switch.
- Copy the running configuration to the startup configuration.

Step 4: Test connectivity.

Verify that the PC hosts can ping one another. Note: It may be necessary to disable the PCs firewall to ping between PCs.

PC1 can ping PC5

PC2 can Ping PC6

PC3 can ping PC7

PC4 can ping PC8

Pings to PCs with others fail.

Create VLANs on the Switches

Create the same VLANs on S2 and s3

Procedure:

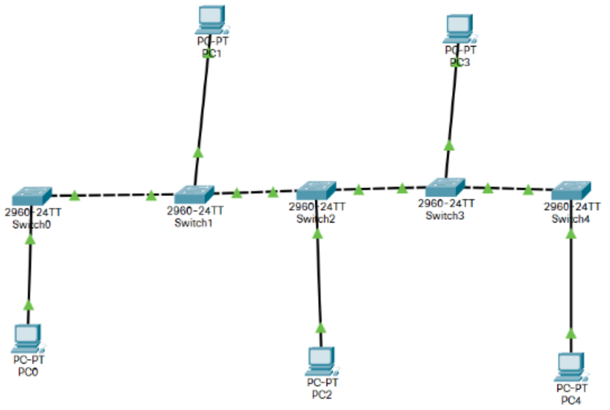
- Let us take 8 p.c's and 2 switches.
- Now connect the 4 pcs to 1 switch and connect the another 4 pcs to another switch through copper straight -Through connection.
- Now connect the both switches with the copper cross- Over connection as shown.
- Designing part is over
- Now we should the configure the pcs. Then we consider 4pcs of one lan and other 4pcs of another lan(therefore each switch has two separate lan connection as shown)
- I configure pcs in this format switch-1 (pc0-10.10.10.1, pc1-10.10.10.2, pc2-10.10.20.1, pc3-10.10.20.2) switch-2 (pc4-10.10.10.3, pc5-10.10.10.4, pc6-10.10.20.3, pc7-10.10.20.4)
- Now the connection part and configuration of pcs are done, now must configure the switches
- As configure the switches using the above lines of code.

Lab 4: Design of Topologies: Bus, Mesh, Ring, Tree and Star topologies

a) Bus Topology

Objectives:

1) Design a BUS Topology using switches with PCs.
2) Verify the connectivity.



Addressing Table:

Device	Interface	IP Address	Subnet Mask
PC0	NIC	192.172.16.1	255.255.255.0
PC1	NIC	192.172.16.2	255.255.255.0
PC2	NIC	192.172.16.3	255.255.255.0
PC3	NIC	192.172.16.4	255.255.255.0
PC4	NIC	192.172.16.5	255.255.255.0

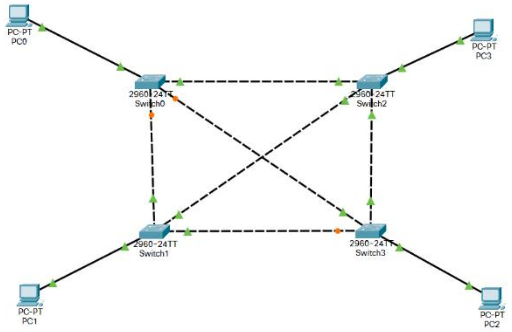
Procedure:

- Consider 5 PC's and 5 Switches
- Connect them as shown in figure.
- No configure the each PC as shown in table.
- Now send the packet from one end to the other node of the terminal.

b) Mesh Topology

Objectives:

1) Design a Mesh Topology using switches with PCs.
2) Verify the connectivity.



Addressing Table:

Device	Interface	IP Address	Subnet Mask
PC0	NIC	10.10.10.1	255.0.0.0
PC1	NIC	10.10.10.2	255.0.0.0
PC2	NIC	10.10.10.3	255.0.0.0
PC3	NIC	10.10.10.4	255.0.0.0
PC4	NIC	10.10.10.5	255.0.0.0

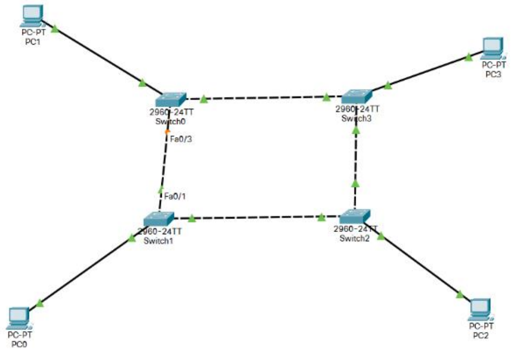
Procedure:

- Consider 5 PC’s and 5 Switches
- Connect them as shown in figure.
- No configure each PC as shown in addressing table.
- Now send the packet from one end to the other node of the terminal.

c) Ring Topology

Objectives:

1) Design a Ring Topology using switches with PCs.
2) Verify the connectivity.



Addressing Table:

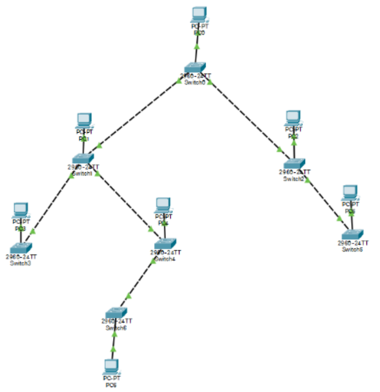
Device	Interface	IP Address	Subnet Mask
PC0	NIC	10.10.10.1	255.0.0.0

PC1	NIC	10.10.10.2	255.0.0.0
PC2	NIC	10.10.10.3	255.0.0.0
PC3	NIC	10.10.10.4	255.0.0.0
PC4	NIC	10.10.10.5	255.0.0.0

Procedure:

- Consider 5 PC's and 5 Switches
- Connect them as shown in figure.
- No configure each PC as shown in addressing table.
- Now send the packet from one end to the other node of the terminal.

d) Tree Topology



Objectives:

1) Design a Tree Topology using switches with PCs.
2) Verify the connectivity.

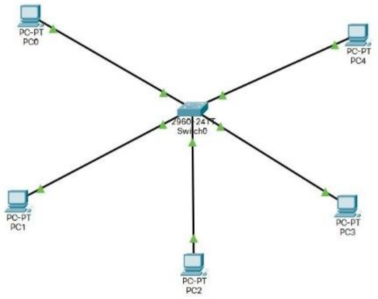
Addressing Table:

Device	Interface	IP Address	Subnet Mask
PC0	NIC	10.10.10.1	255.0.0.0
PC1	NIC	10.10.10.2	255.0.0.0
PC2	NIC	10.10.10.3	255.0.0.0
PC3	NIC	10.10.10.4	255.0.0.0
PC4	NIC	10.10.10.5	255.0.0.0
PC5	NIC	10.10.10.6	255.0.0.0
PC6	NIC	10.10.10.7	255.0.0.0
PC7	NIC	10.10.10.8	255.0.0.0

Procedure:

- Consider 7 PC's and 7 Switches
- Connect them as shown in figure.
- No configure each PC as shown in addressing table.
- Now send the packet from one end to the other node of the terminal.

e) Star Topology



Objectives:

1) Design a Star Topology using switches with PCs.
2) Verify the connectivity.

Addressing Table:

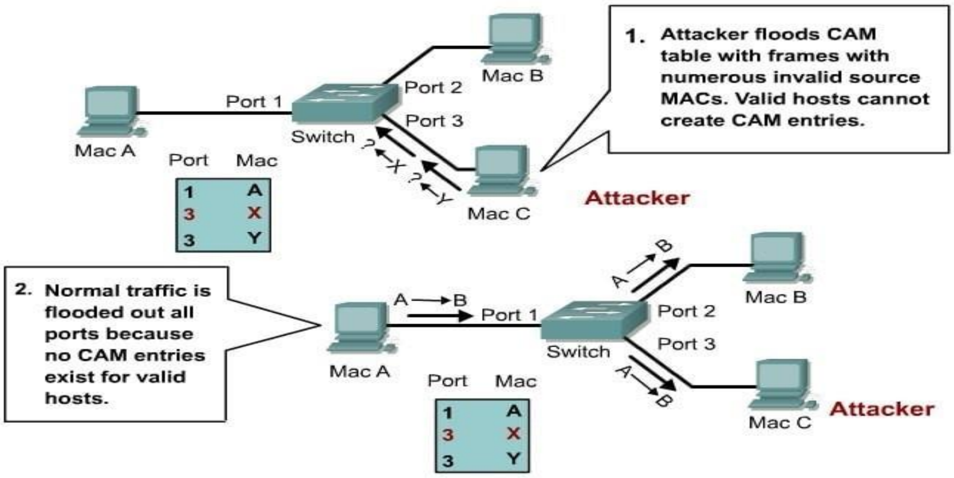
Device	Interface	IP Address	Subnet Mask
PC0	NIC	10.10.10.1	255.0.0.0
PC1	NIC	10.10.10.2	255.0.0.0
PC2	NIC	10.10.10.3	255.0.0.0
PC3	NIC	10.10.10.4	255.0.0.0
PC4	NIC	10.10.10.5	255.0.0.0

Procedure:

- Consider 5 PC’s and 1 Switches
- Connect them as shown in figure.
- No configure each PC as shown in addressing table.
- Now send the packet from one end to the other node of the terminal.

LAB 5: Port-Security Configuration to prevent MAC flooding attack

Switch port Security is a **network security** feature that associates specific MAC addresses of devices (such as PCs) with specific interfaces on a switch. This will enable you to restrict access to a given switch interface so that only the authorized devices can use it. If an unauthorized device is connected to the same port, you can define the action that the switch will take, such as discarding the traffic, sending an alert, or shutting down the port.



Now let's configure port security in Packet Tracer.

1. Build the network topology:

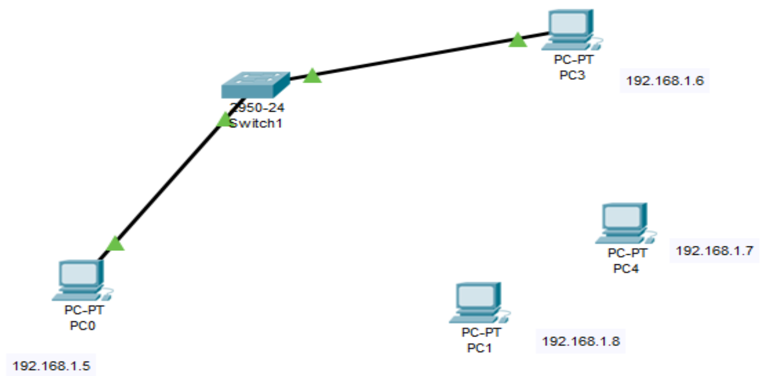
PC1 connects to fa0/1 and PC2 to fa0/2 of the switch

2. Now configure switch port security on switch interfaces.

We'll configure port security interfaces on fa0/1 and fa0/2.

To do this, we'll:

2. Configure the port as an **access port**
3. Enable **port security**
4. Define which **MAC addresses** are allowed to send frames through this interface.



The *sticky* keyword instructs the switch to **dynamically** learn the MAC address of the currently connected host.

You can add these two optional commands.

- defining the action that the switch will take when a frame from an unathorized device is received. This is done using the switchport port-security violation {protect | restrict | shutdown}interface command. All three options discard the traffic from the unauthorized device.

- defining the maximum number of MAC addresses that can be received on the port using the switchport port-security maximum NUMBER interface submode comman

Port Security Violation Modes:

Security Violation Modes

Violation Mode	Forwards Traffic	Sends Syslog Message	Displays Error Message	Increases Violation Counter	Shuts Down Port
Protect	No	No	No	No	No
Restrict	No	Yes	No	Yes	No
Shutdown	No	No	No	Yes	Yes

switchport port security example

8

In our topology **PC0** is connected with **F0/1** port of switch. Enter following commands to secure **F0/1** port

- Switch>enable
- Switch#configure terminal
- Switch(config)#interface fastethernet 0/1
- Switch(config-if)#switchport mode access
- Switch(config-if)#switchport port-security
- Switch(config-if)#switchport port-security maximum 1
- Switch(config-if)#switchport port-security violation shutdown
- Switch(config-if)#switchport port-security mac-address sticky
- Switch(config-if)#ex
- Switch(config)#ex

Java

Networks and Communication Department

```
Switch# Show port-security
Switch# sh mac-address-table
Switch# sh ip int brief
```

Procedure:

- Take 4 pcs and a single switch.
- Now configure the three pcs as the same series of IP address.
- Now we just connect the two pcs to the switch because we are here checking the shutdown case.
- After connecting the two pcs we just CLI the switch as shown
- Now we just change the connections to the another pc of which is connected to the fa0/1 to another pc
- Now we send a packet for the cmd prompt of pc0 to pc1 so that we get some TTL value.
- Now we send a packet for the cmd prompt of pc1 to pc2 so that we get some TTL value.
- Again we do the same thing by changing another pc to connect to the same switch port. But in this case the ping the pc will get the request timed out.pc3 to pc2
- Her we can now that switch not accepting 3 mac address to the same port of the switch.
- So finally the switch is shutdown.

Lab 6: Router Serial Point to Point Connection HDLC, PPP with PAP and CHAP

HDLC (High-level Data Link Control) is a WAN protocol intended to perform the encapsulation of the data in the data link layer. The encapsulation of the data means to change the format of the of the data. HDLC protocol is developed by IBM and submitted to the ANSI and ISO for the acceptance as the international standards.

HDLC has two versions. One of them is the standard one and the other is the Cisco proprietary version. The frame of standard version and Cisco proprietary version is similar. Only in Cisco proprietary HDLC, there is one additional proprietary field. Below, you can check both of the frames:

Cisco HDLC is the default enabled WAN protocol of Cisco for Point-to-Point WAN links. And we can use Cisco HDLC only between Cisco devices. Other vendor devices cannot use Cisco HDLC.

Lastly, there is no Authentication mechanism in HDLC. So, security is a concern for this WAN protocol.

PPP (Point to Point Protocol) is also a WAN Encapsulation Protocol that is based on HDLC but we can say that it is the enhanced version of HDLC. There are many additional features in **PPP** if we compare with HDLC.

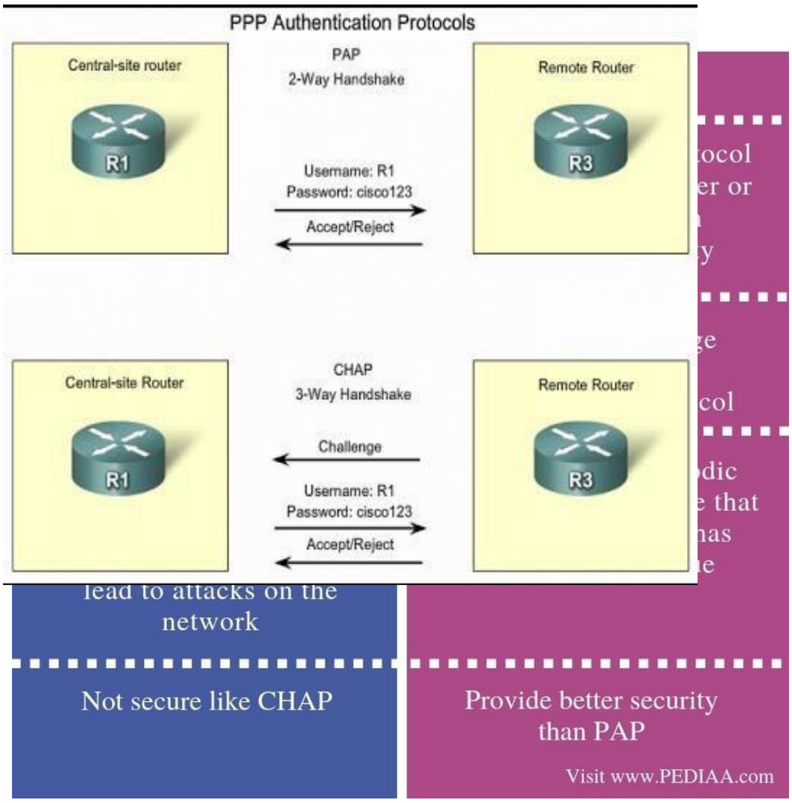
PPP supports two Authentication Protocols. These Authentication Protocols are:

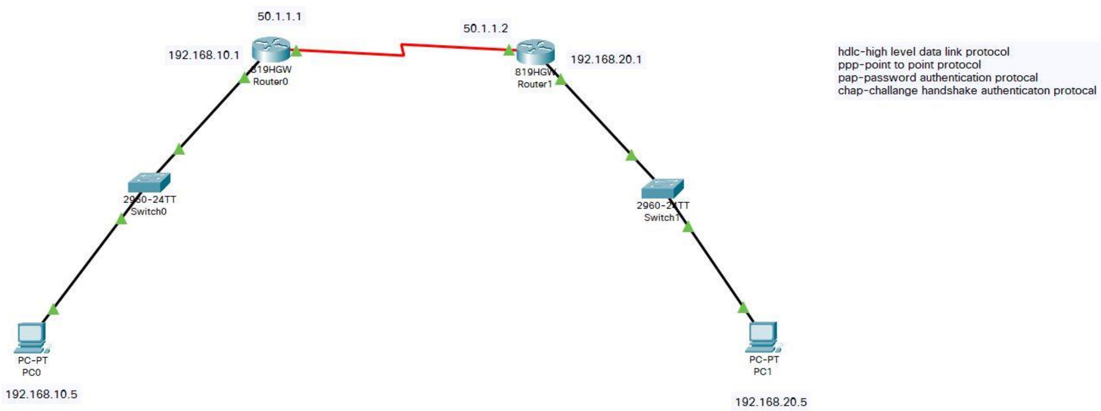
- PAP (Password Authentication Protocol)
- **CHAP (Challenge Handshake Authentication Protocol)**

PAP (Password Authentication Protocol) is the simplest Authentication method. It uses 2- way handshake. Both ends send the passwords in "**clear text**" in this method. And passwords are exchanged only at the beginning.

CHAP (Challenge Handshake Authentication Protocol) is the more complex Authentications method. CHAP uses 3-way handshake and with this mechanism it checks the remote node periodically. CHAP uses MD5 hash. One end sends "**Hash**" to other node and the other node also sends a hash. If the hashes are same, then the communication starts.

PAP v/s CHAP





By default, HDLC encapsulation is works in CISCO devices. In case it is not there Then by simple command we can configure:

Addressing Table:

Device	Interface	IP Address	Subnet Mask
PC0	NIC	192.168.10.5	255.255.255.0
PC1	NIC	192.168.20.5	255.255.255.0
Router1 Fast Ethernet 0/1	NIC	192.168.10.1	255.255.255.0
Router1 Serial 0/0	NIC	50.1.1.1	255.0.0.0
Router2 Fast Ethernet 0/1	NIC	192.168.20.1	255.255.255.0
Router2 Serial 0/0	NIC	50.1.1.2	255.0.0.0

Objectives:

1. Design a WAN with 2 Routers
2. Configure the Routers
3. Apply PPP (PAP and CHAP) authentication on it.
4. Verify Connection.

Router Configuration :

Router> enable

```
Router# configure terminal
Router(config)# interface gigabitethernet 0/1
Router(config-if)# ip address 192.168.1.2
255.255.255.0
Router(config)# ip route 192.168.1.0 255.255.0.0
192.168.2.2 Router(config-if)# no shutdown
```

PROCEDURE:

- Here we need 2 PC's, 2 Routers, 2 Switches we need to connect them as shown in the figure.
- After connecting them we need to configure both the pc with different IP address.
- Here both the routers are connected in serial and also the ip address of routers are also to be different (50.1.1.1, 50.1.1.2)
- We need to assign t hip address as shown in the table
- We configure the routers with
- After this need to change both the routers ppp encapsulation mode.
- In ppp encapsulation mode we can get access of both pap, chap authentication to enhance.
- The router configuration for PAP and CHAP is

Pap(Router0):

```
Router(config-if)#en
Router(config-if)#encapsulation ppp
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0, changed state
to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0, changed state
to up

Router(config-if)#host
Router(config-if)#exit
Router(config)#hos
Router(config)#hostname R1
R1(config)#wr
           ^
% Invalid input detected at '^' marker.

R1(config)#exit
R1#wr
Building configuration...
[OK]
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#enable password CISCO
R1(config)#ex
R1#wr
Building configuration...
[OK]
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int se0
```

```
R1(config-if)#ppp au
R1(config-if)#ppp authentication pap
R1(config-if)#
%SYS-5-CONFIG_I: Configured from console by console

%SYS-5-CONFIG_I: Configured from console by console

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0, changed state
to down

R1(config-if)#exit
R1(config)#user
R1(config)#username R2 password CISCO
R1(config)#int se0
R1(config-if)#ppp pap se
R1(config-if)#ppp pap sent-username R1 password SCOPE
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0, changed state
to up
```

Pap(Router1):

```
Router(config-if)#en
Router(config-if)#encapsulation ppp
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0, changed state
to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0, changed state
to up

Router(config-if)#host
Router(config-if)#exit
Router(config)#hos
Router(config)#hostname R2
R2(config)#wr
           ^
% Invalid input detected at '^' marker.

R2(config)#exit
R2#wr
Building configuration...
[OK]
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#enable password CISCO
R2(config)#ex
R2#wr
Building configuration...
[OK]
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#int se0
R2(config-if)#ppp au
R2(config-if)#ppp authentication pap
R2(config-if)#
%SYS-5-CONFIG_I: Configured from console by console

%SYS-5-CONFIG_I: Configured from console by console

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0, changed state
to down

R2(config-if)#exit
R2(config)#user
R2(config)#username R1 password SCOPE
R2(config)#int se0
R2(config-if)#ppp pap se
R2(config-if)#ppp pap sent-username R2 password CISCO
R2(config-if)#
R2(config-if)#
R2(config-if)#
```

```
R2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0, changed state
to up
```

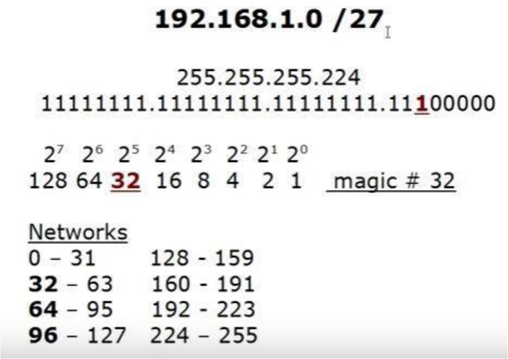
Chap(Router0)R1:

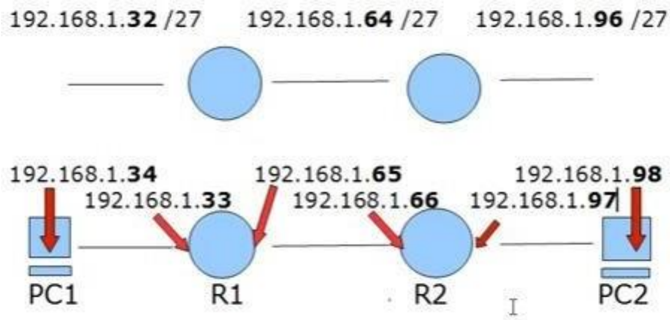
```
R1>en
R1>enable
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#us
R1(config)#username R2 pass
R1(config)#username R2 password CISCO
R1(config)#int se0
R1(config-if)#ppp au
R1(config-if)#ppp authentication chap
R1(config-if)#
%LINK-3-UPDOWN: Interface Serial0, changed state to down

%LINK-5-CHANGED: Interface Serial0, changed state to up
```

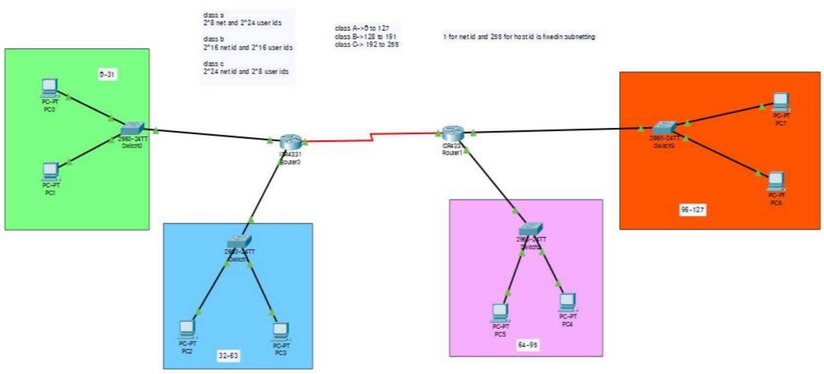
Chap(Router1)R2:

```
R2>en
R2>enable
Password:
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#us
R2(config)#username R1 pass
R2(config)#username R1 password SCOPE
R2(config)#int se0
R2(config-if)#ppp au
R2(config-if)#ppp authentication chap
R2(config-if)#
%LINK-3-UPDOWN: Interface Serial0, changed state to down

%LINK-5-CHANGED: Interface Serial0, changed state to up
```

Lab 7: Design a network using Subnetting for a class C Ipv4 address and Configure it in Router.







Objectives:

1. Design Subnet network using two routers and switches with two PCs each.
2. Configure Subnet
3. Verify the connectivity.

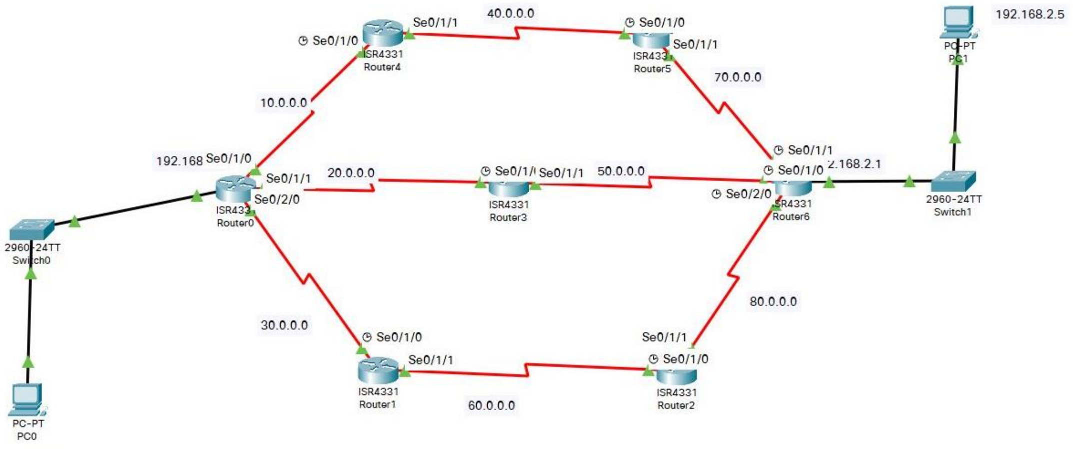
Addressing Table:

Device	Interface	IP Address	Subnet Mask
PC0	NIC	192.168.1.2	255.255.255.224
PC1	NIC	192.168.1.2	255.255.255.224
PC2	NIC	192.168.1.34	255.255.255.224
PC3	NIC	192.168.1.35	255.255.255.224
Router 0 gi0/0/0	NIC	192.168.1.1	255.255.255.224
Router 0 gi0/0/1	NIC	192.168.1.33	255.255.255.224
PC4	NIC	192.168.1.67	255.255.255.224
PC5	NIC	192.168.1.66	255.255.255.224
PC6	NIC	192.168.1.98	255.255.255.224
PC7	NIC	192.168.1.99	255.255.255.224
Router 1 gi0/0/0	NIC	192.168.1.65	255.255.255.224
Router 1 gi0/0/1	NIC	192.168.1.97	255.255.255.224
Router 1 Serial 0/2	NIC	192.168.1.129	255.255.255.224
Router 2 Serial 0/2	NIC	192.168.1.130	255.255.255.224

Procedures:

- 4 switches 8PC's and 2 Routers are connected as follows.
- For each 2 pcs will be in one subnetting configuration
- The 1st number and last number for each subnet is reserved for host id
- We can use in b/w them such that for ex-0-31 in this we can access only 1-30 address.
- Now we need to configure each pc as shown in the table.
- Now we need to configure both the routers.

Lab 8: Design and configure a network using RIP Routing protocol.



Objectives:

1. Design a network using seven routers and switches with one source and one destination PC
2. Configure all the routers with RIP routing.
3. Verify the connectivity.

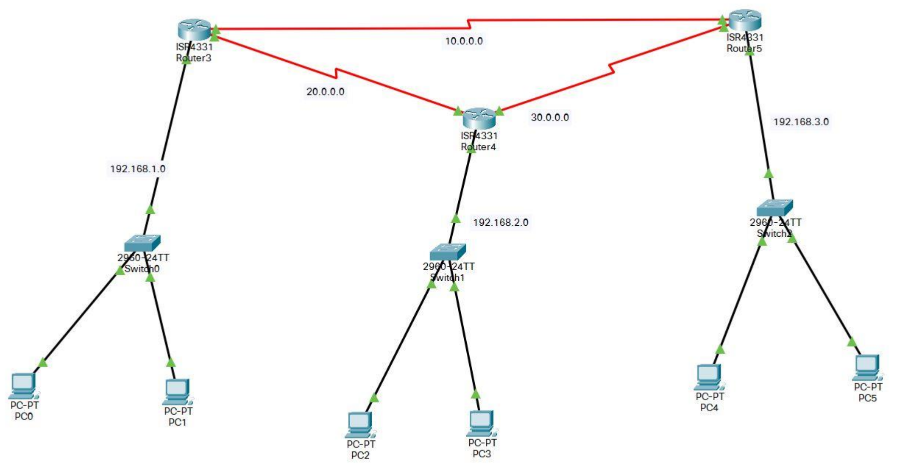
Addressing Table:

Device	Interface	IP Address	Subnet Mask
PC1	NIC	192.168.1.5	255.255.255.0
PC2	NIC	192.168.2.5	255.255.255.0
Router 0 gi 0/0/0	NIC	192.168.1.1	255.255.255.0
Router 0 Serial 0/1/0	NIC	10.0.0.1	255.0.0.0
Router 0 Serial 0/1/1	NIC	20.0.0.1	255.0.0.0
Router 0 Serial 0/2/0	NIC	30.0.0.1	255.0.0.0
Router 1 Serial 0/1/0	NIC	30.0.0.2	255.0.0.0
Router 1 Serial 0/1/1	NIC	60.0.0.1	255.0.0.0
Router 2 Serial 0/1/0	NIC	60.0.0.2	255.0.0.0
Router 2 Serial 0/1/1	NIC	80.0.0.1	255.0.0.0
Router 3 Serial 0/1/0	NIC	20.0.0.2	255.0.0.0
Router 3 Serial 0/1/1	NIC	50.0.0.1	255.0.0.0
Router 4 Serial 0/1/0	NIC	10.0.0.2	255.0.0.0
Router 4 Serial 0/1/1	NIC	40.0.0.1	255.0.0.0
Router 5 Serial 0/1/0	NIC	40.0.0.2	255.0.0.0
Router 5 Serial 0/1/1	NIC	70.0.0.1	255.0.0.0
Router 6 Serial 0/1/0	NIC	50.0.0.2	255.0.0.0
Router 6 Serial 0/1/1	NIC	70.0.0.2	255.0.0.0
Router 6 Serial 0/2/0	NIC	80.0.0.2	255.0.0.0
Router 6 gi 0/0/0	NIC	192.168.2.1	255.255.255.0

PROCEDURE:

- Take 7 routers, 2 switches and 2 pcs
- Take minimum of 3 serial ports in each router and then we need to configure them.
- Now we configure the pcs and routers as per the table
- After configuration of each router and PC's
- Now we need to configure the routers each other to send the packet
- But the packets wont be sent since we are in the rip routing mode we need to give a command to access the rip routing
- For each router we need to configure in the configuration mode and
    - Router rip
    - Network ******(where star represents the ip series of the router on which to which we need to transfer)
- For ex., router 0 scenario be like
    - Router rip
    - Network 10.0.0.0
    - Network 20.0.0.0
    - Network 30.0.0.0
    - Network 192.168.1.0

Lab 9: Design and configure a network using OSPF Routing protocol.



Objectives:

1. Design a network using three routers and switches with two PCs each.
2. Configure all the routers with OSPF routing.
3. Verify the connectivity.

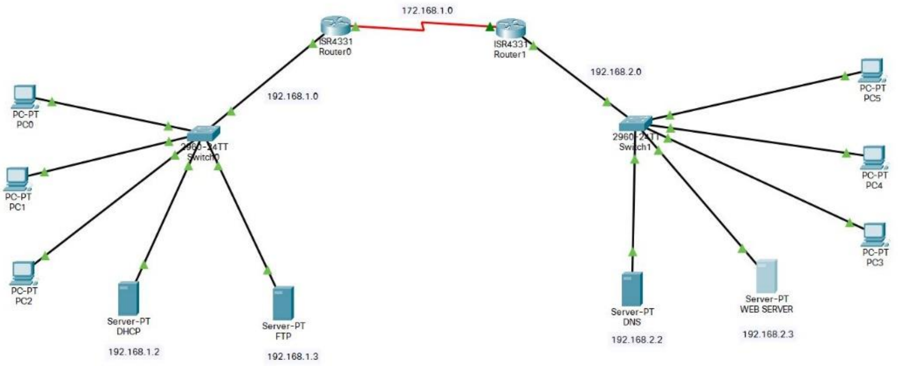
Addressing Table:

Device	Area	IP Address	wild card bits	Subnet Mask
PC0	0	192.168.1.2		255.255.255.0
PC1	0	192.168.1.3		255.255.255.0
PC2	0	192.168.2.2		255.255.255.0
PC3	0	192.168.2.3		255.255.255.0
PC4	0	192.168.3.2		255.255.255.0
PC5	0	192.168.3.3		255.255.255.0
Router 3 gi0/0/0	0	192.168.1.1	0.0.0.255	255.255.255.0
Router 3 Serial 0/1/0	0	10.0.0.1	0.255.255.255	255.0.0.0
Router 3 Serial 0/1/1	0	20.0.0.1	0.255.255.255	255.0.0.0
Router 4 gi0/0/0	0	192.168.2.1	0.0.0.255	255.255.255.0
Router 4 Serial 0/1/0	0	20.0.0.2	0.255.255.255	255.0.0.0
Router 4 Serial 0/1/1	0	30.0.0.1	0.255.255.255	255.0.0.0
Router 5 gi0/0/0	0	192.168.3.1	0.0.0.255	255.255.255.0
Router 5 Serial 0/1/0	0	10.0.0.2	0.255.255.255	255.0.0.0
Router 5 Serial 0/1/1	0	30.0.0.2	0.255.255.255	255.0.0.0

Procedure:

- Take 3 routers 3 switches 6 PC’s
- Now connect the shown in the figure (for adding serial in each router we need to off the router and add the serial component on the router again)
- after this the designing will be done
- Now we configure each Router, PC’s as per the connections
    - For each switch we need to allot a certain series as shown in the table
    - Now for router we need to give gateway address and also we need to note this address in the in the PC’s
    - Now configure the router as per the table
- After completion of these now we need to go to CLI on each router and give cmds as shown in top for their respective router

Lab 10: Design and configure a network using DHCP, DNS, Web Server and FTP protocol.



Objectives:

1. Design a network using two routers and switches with 3 - 4 PCs each and one DHCP, DNS, Web, and FTP server in the network.
2. Configure all the routers with any routing scheme.
3. Configure all the servers.
4. Verify the servers working.

Addressing Table:

Device	IP Address	Subnet Mask
PC0		
PC1		
PC2		
PC3		
PC4		
PC5		
DHCP Server	192.168.1.2	255.255.255.0
FTP Server	192.168.1.3	255.255.255.0
DNS Server	192.168.2.2	255.255.255.0
Web/HTTP Server	192.168.2.3	255.255.255.0
Router 1 giga0/0/0	192.168.1.1	255.255.255.0
Router 1 Serial 0/1/0	172.168.1.1	255.255.0.0
Router 2 giga0/0/0	192.168.2.1	255.255.255.0
Router 2 Serial 0/1/0	172.168.1.2	255.255.0.0

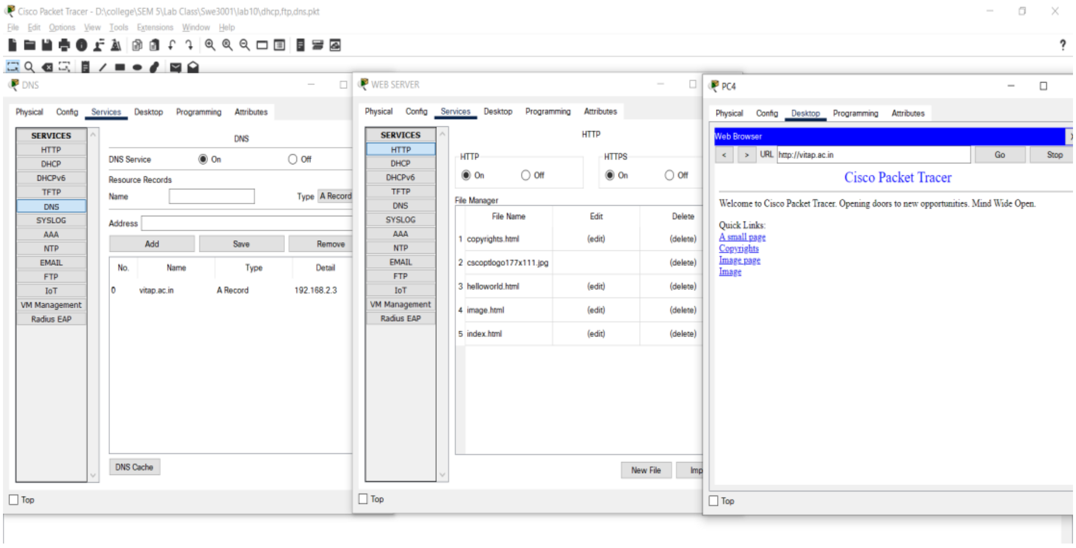
<u>Procedure:</u>

- Take 2 routers 2 switches 6 PC’s and 4 servers. connect them as shown in fig.
- Now the designing part is over. Now we need to configure them
- Now by configuring DHCP our monitors will be configured
- Now by configuring ftp server by keeping a new username and password we can access the files in FTP can be accessed through the same username and

password and also we can see the file name of what are been present in the FTP server

- Now we configure the DNS server by adding a name, type, detail.
- Now open any pc and go to web browser and type a website present in DNS server it will display the site with the help of HTTP server.

DNS & HTTP



- As in HTTP the index code is defined for cisco packet tracer so it is displaying the index for vitap.ac.in website
- As we mentioned above the DNS server is sending the website address to webserver and then it is going for display in the PC
- Now we configure the webserver by editing the files present in it or by accessing the files in the webserver in any computer
- Consider a computer opposite to FTP sever and try to access the FTP server if the computer terminal ask for username and password then we are able to access the files in FTP server

FTP: